

# DSA Day 8: Doubly Linked List

Book: Data Structures Through C in Depth - Ch3 §3.2 · pp 63-72 | Code in Python

## Overview

Each node has prev and next, enabling backward traversal and  $O(1)$  deletion when you hold the node.

## Key concepts to master

- Two links per node
- Forward & backward traversal
- Insert/delete update up to 4 links
- Reversal

## Python implementation

```
class DNode:
    def __init__(self,d): self.data=d; self.prev=None; self.next=None
def insert_after(p,v):
    n=DNode(v); n.prev=p; n.next=p.next
    if p.next: p.next.prev=n
    p.next=n
```

## Complexity

Insert/delete  $O(1)$  at a node; traversal  $O(n)$ .

## Common pitfalls

- Forgetting to fix prev links
- Edge cases at head/tail

## Practice

- Delete the middle node
- Print forward and backward

